# AP Calc Lesson 2.3 Homework

**Answer Key**

1. a. $g(4) = \lim_{x \to 4^-} g(x) = \lim_{x \to 4^+} h(x)$

   b. Part a must be true. Additionally, $\lim_{x \to 4^-} f'(x) = \lim_{x \to 4^+} f'(x)$. This can be verified by showing $\lim_{x \to 4^-} g'(x) = \lim_{x \to 4^+} h'(x)$.

2. $A'(3) = 145.149$ attendees per minute

3. a. Although $f$ is continuous at $x = 1$ $\left(\lim_{x \to 1^-} f(x) = f(1) = \lim_{x \to 1^+} f(x) = 7\right)$, $f$ is not differentiable at $x = 1$. If $h(x) = 7$, then function $h$ is a horizontal line and $\lim_{x \to 1^-} f'(x) = \lim_{x \to 1^-} h'(x) = 0$. Since $f$ is a linear function with a slope of $3$ for $x > 1$, $\lim_{x \to 1^+} f'(x) = 3$. Since $\lim_{x \to 1^-} f'(x) \neq \lim_{x \to 1^+} f'(x)$, $f$ is not differentiable at $x = 1$.

   b. $\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} h(x) = 3(1) - 5 = -2.$

   $\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} (3x + 4) = 3(1) + 4 = 7$

   Since $\lim_{x \to 1^-} f(x) \neq \lim_{x \to 1^+} f(x)$, $f$ is not continuous and therefore can not be differentiable at $x = 1$.

   c. Yes, $f$ is differentiable at $x = 1$. Since $\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (x^2 + x + 5) = f(1) = 7$, and $\lim_{x \to 1^+} f(x) = 7$, $f$ is continuous at $x = 1$. Since $\lim_{x \to 1^-} f'(x) = \lim_{x \to 1^-} (2x + 1) = 2(1) + 1 = 3$, and $\lim_{x \to 1^+} f'(x) = 3$, $f$ is also differentiable at $x = 1$.

4. a. $f'(2) = 1.523$, $f'(3)$ is undefined, $f'(4) = -1.523$.

   b. The line tangent to $f$ at $x = 2$ has a positive slope, the graph of $f$ has a sharp turn at $x = 3$, and the line tangent to $f$ at $x = 4$ has a negative slope with the same steepness as the slope at $x = 2$.

5. a. The function $f$ is not differentiable at $x = 2$ because the graph of $f$ has a vertical tangent at $x = 2$. Since the slope of the tangent line is undefined, $f'(2)$ does not exist. At $x = 3$, the graph of $f$ has a corner or cusp, so $f'(3)$ does not exist. The graph of $f$ has a jump discontinuity at $x = 5$, so $f$ is not differentiable at $x = 5$. The function $f$ is not differentiable at $x = 6$ because the graph of $f$ has removable discontinuity at $x = 6$.

   b. $f'(4) = -1$

   c. $f'(1) = 0$

6. Function $h$ is differentiable at $x = 2$ since $\lim_{x \to 2^-} h(x) = h(2) = 4(2) - 7 = 1$ and $\lim_{x \to 2^+} h(x) = f(2) = 1$, which shows that $h$ is continuous at $x = 2$ and $\lim_{x \to 2^-} h'(x) = \lim_{x \to 2^+} h'(x) = 4$.

7. a. $f(\pi) = 7$
   b. $\lim_{x \to \pi^-} f'(x) = 2.5$
   c. $c = 2.5, d = 7 - 2.5\pi \approx -0.854$

8. a. $x = -2$ and $x = 1$. Since $f$ is not defined at these locations, $f'$ does not exist, which means the function $f$ is not differentiable in these two places.
   b. $f'(-1) < f'(-3) < f'(3)$
   c. $f'(2) = 3$

9. A

10. D